

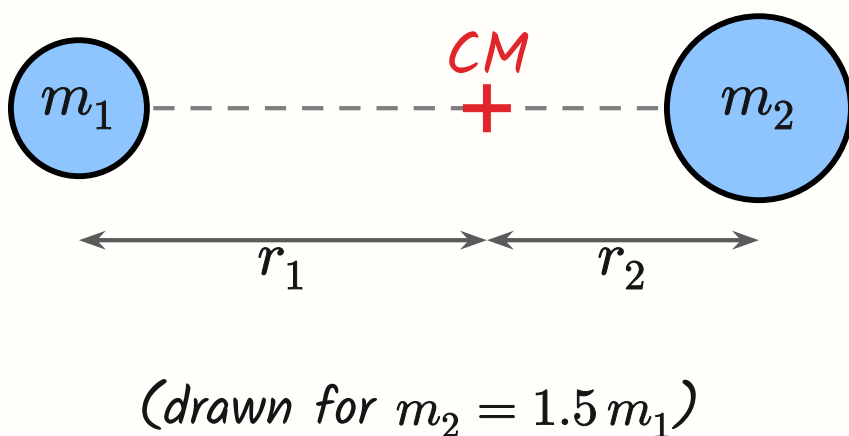
Centre of Mass

1 System of particles

$$\vec{r}_{cm} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots}{m_1 + m_2 + \dots} = \frac{\sum m_i \vec{r}_i}{M}$$

$x_{cm} = \sum m_i x_i / M$ (same for y, z) $\cdot M = \sum m_i$ in kg $\cdot \vec{r}$ in m

2 Two particles, separation d



- $m_1 r_1 = m_2 r_2$: CM is nearer the heavier mass.
- $r_1 = \frac{m_2 d}{m_1 + m_2}$, $r_2 = \frac{m_1 d}{m_1 + m_2}$
- Equal masses: CM at the midpoint.

Example

2 kg at $x = 0$, 3 kg at $x = 5$ m: $x_{cm} = \frac{2(0) + 3(5)}{5} = 3$ m

Golden points

- ★ CM need not lie on the body (ring, bangle, L-shape).
- ★ Uniform body with symmetry: CM lies on the symmetry axis.
- ★ Its place in the body does not depend on the origin chosen.

Summary

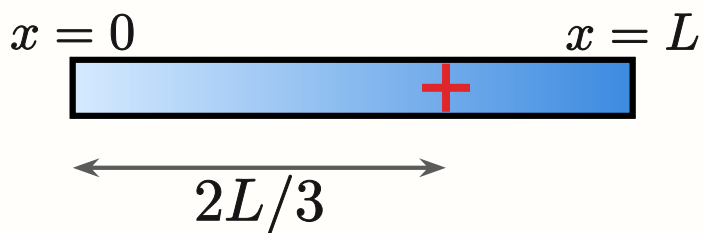
$$\vec{r}_{cm} = \sum m_i \vec{r}_i / M \quad \bullet \quad m_1 r_1 = m_2 r_2 \quad \bullet \quad r_1 = \frac{m_2 d}{m_1 + m_2}$$

CM of Continuous Bodies

1 Integral form

$$\vec{r}_{cm} = \frac{1}{M} \int \vec{r} dm$$

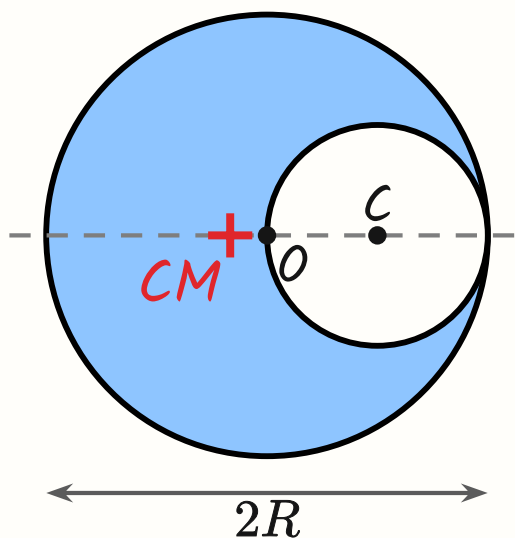
$$dm = \lambda dx \text{ (rod)} \cdot \sigma dA \text{ (plate)} \cdot \rho dV \text{ (solid)}$$



Example: rod with $\lambda = kx$

$$x_{cm} = \frac{\int_0^L kx^2 dx}{\int_0^L kx dx} = \frac{2L}{3}, \text{ not } \frac{L}{2}$$

2 Cavity = negative mass



$$x_{cm} = \frac{M_1 x_1 - M_2 x_2}{M_1 - M_2}$$

M_2 : removed part · plate: $M \propto \text{area}$

Example

Disc R , hole $R/2$ touching O :

$$M_2 = M_1/4, x_2 = R/2 \Rightarrow x_{cm} = -R/6.$$

Golden points

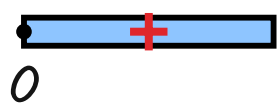
★ $CM = CG$ only if g is uniform; a hole pushes the CM away from it.

Summary

$$\vec{r}_{cm} = \frac{1}{M} \int \vec{r} dm \quad \bullet \quad x_{cm} = \frac{M_1 x_1 - M_2 x_2}{M_1 - M_2}$$

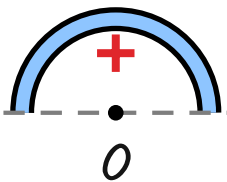
CM of Standard Bodies

1 Uniform bodies, distance of CM from O



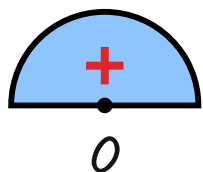
Rod, length L

$\frac{L}{2}$ from an end



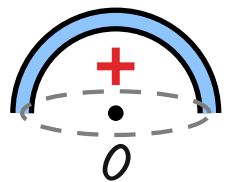
Semicircular ring (wire)

$\frac{2R}{\pi}$



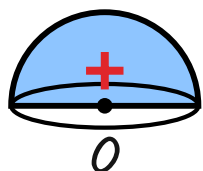
Semicircular disc (plate)

$\frac{4R}{3\pi}$



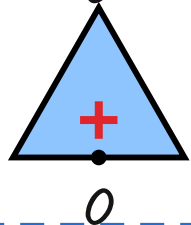
Hollow hemisphere (shell)

$\frac{R}{2}$



Solid hemisphere

$\frac{3R}{8}$



Solid cone, height h

$\frac{h}{4}$ above the base

Golden points

★ O = centre of the full circle / flat face; distance along the symmetry axis.

★ Trap: shell $R/2$ but solid hemisphere $3R/8$; ring $2R/\pi$ but disc $4R/3\pi$.

Summary

ring $\frac{2R}{\pi}$ • disc $\frac{4R}{3\pi}$ • shell $\frac{R}{2}$ • solid $\frac{3R}{8}$ • cone $\frac{h}{4}$

Motion of the CM

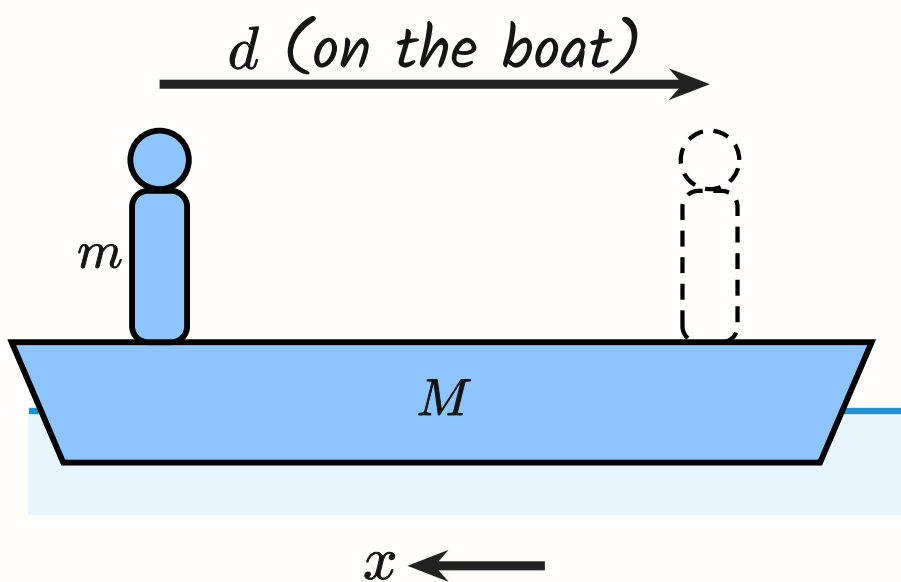
1 Newton's second law for a system

$$\vec{v}_{cm} = \frac{\sum m_i \vec{v}_i}{M}, \quad \vec{P} = M\vec{v}_{cm}$$

$$M\vec{a}_{cm} = \vec{F}_{ext} = \frac{d\vec{P}}{dt}$$

$\vec{P}$ in kg m s^{-1} · internal forces cancel in pairs (Newton's 3rd law)

2 $\vec{F}_{ext} = 0$: man walking on a boat



$$x = \frac{m d}{m + M}$$

starts at rest · no friction from water · boat moves back by x

Example

$m = 60 \text{ kg}$, $M = 240 \text{ kg}$,
 $d = 4 \text{ m}$: $x = 0.8 \text{ m}$; man
 moves 3.2 m w.r.t. ground.

Golden points

- ★ $\vec{F}_{ext} = 0 \Rightarrow \vec{P}$, $\vec{v}_{cm}$ constant: a CM at rest stays at rest.
- ★ Momentum is conserved along any direction in which $F_{ext} = 0$.

Summary

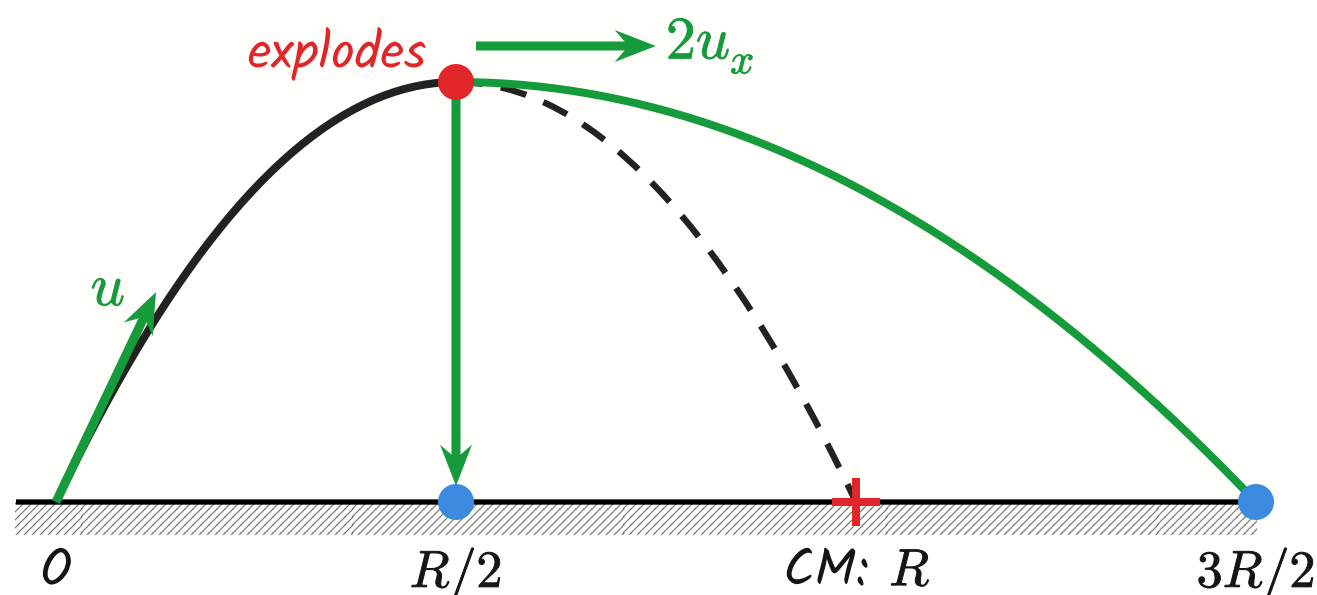
$$M\vec{a}_{cm} = \vec{F}_{ext} \quad \bullet \quad \vec{P} = M\vec{v}_{cm} \quad \bullet \quad \vec{F}_{ext} = 0 \Rightarrow \vec{v}_{cm} \text{ const}$$

Explosion of a Projectile

1 Only gravity acts on the system

$M\vec{a}_{cm} = M\vec{g} \Rightarrow$ CM keeps the same parabola
explosion forces are internal • valid until a fragment hits the ground

2 Two equal halves at the top



- One half stops and falls straight down: lands at $R/2$.
- Momentum along x : $mu_x = \frac{m}{2}(0) + \frac{m}{2}v \Rightarrow v = 2u_x$.
- Level ground: $x_{cm} = \frac{R/2 + x_2}{2} = R \Rightarrow x_2 = \frac{3}{2}R$.

Golden points

- ★ Momentum is conserved in the explosion; KE is NOT (chemical energy is released).
- ★ Gravity's impulse during the short explosion is negligible.

Summary

$\vec{a}_{cm} = \vec{g}$ • CM lands at R • other half at $\frac{3}{2}R$

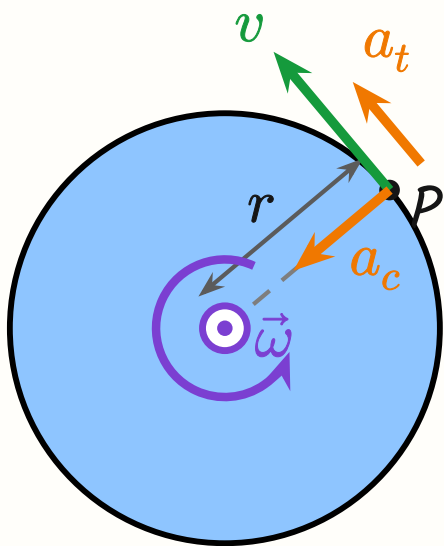
Rotational Kinematics

1 Rigid body about a fixed axis

$$\omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

rad, rad s^{-1} , rad s^{-2} • $N \text{ rpm}$: $\omega = 2\pi N/60$

2 Point P at distance r



- $\vec{v} = \vec{\omega} \times \vec{r}$, $v = \omega r$ (tangent)
- $a_t = \alpha r$ (tangent)
- $a_c = \omega^2 r$ (towards the axis)
- Anticlockwise: $\vec{\omega}$ out of page

3 Equations: constant α only

$$\omega = \omega_0 + \alpha t \quad \theta = \omega_0 t + \frac{1}{2}\alpha t^2 \quad \omega^2 = \omega_0^2 + 2\alpha\theta$$

Example

From rest, $\alpha = 2 \text{ rad s}^{-2}$, 5 s: $\omega = 10 \text{ rad s}^{-1}$, $\theta = 25 \text{ rad} \approx 3.98 \text{ rev}$

Golden points

- ★ Same θ, ω, α for all points; v, a grow with r .
- ★ Uniform rotation: $a_t = 0$ but $a_c = \omega^2 r \neq 0$.

Summary

$$v = \omega r \quad \bullet \quad a_t = \alpha r \quad \bullet \quad a_c = \omega^2 r \quad \bullet \quad \omega^2 = \omega_0^2 + 2\alpha\theta$$

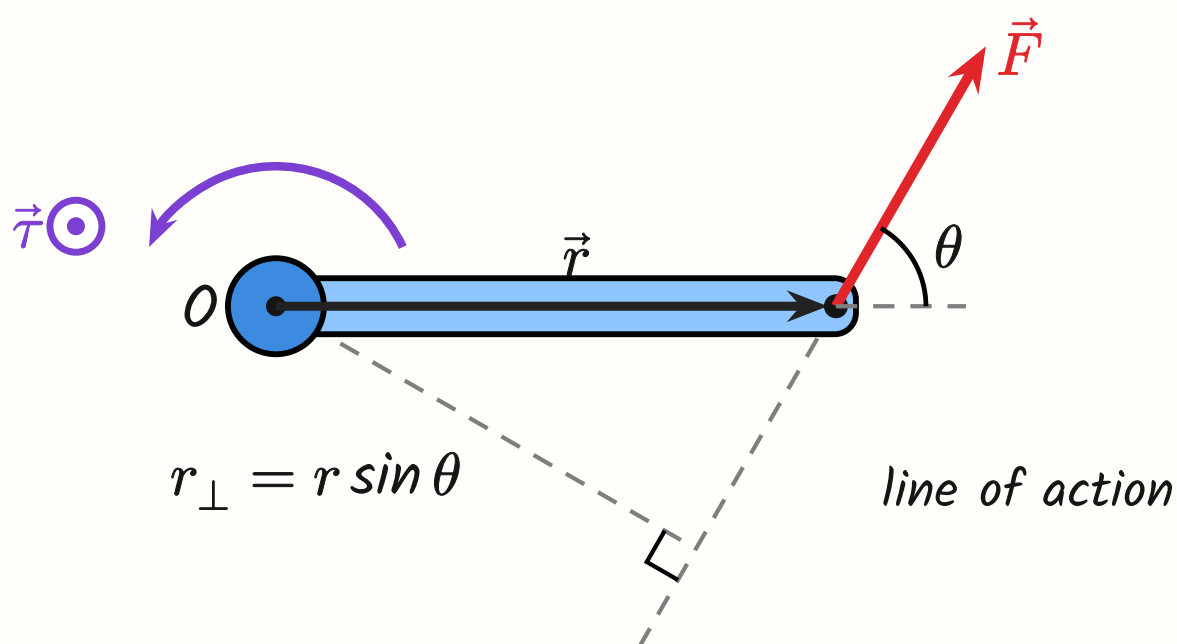
Torque

1 Turning effect of a force

$$\vec{\tau} = \vec{r} \times \vec{F}, \quad \tau = rF \sin \theta = r_{\perp} F$$

$\vec{r}$: O to the point where $\vec{F}$ acts $\cdot$ Nm (never J, though $[\text{ML}^2\text{T}^{-2}]$)

2 Moment arm and direction



- $r_{\perp}$ = perpendicular distance from O to the line of action of $\vec{F}$.
- Right-hand rule: curl fingers from $\vec{r}$ to $\vec{F}$; thumb gives $\vec{\tau}$ ($\perp$ to both).
- Here the turning is anticlockwise, so $\vec{\tau}$ points out of the page.

Golden points

- ★ $\tau = 0$ when the line of action passes through O ($\theta = 0^\circ, 180^\circ$).
- ★ Torque depends on the point O chosen: always say about which point.

Summary

$$\vec{\tau} = \vec{r} \times \vec{F} \quad \bullet \quad \tau = r_{\perp} F = rF \sin \theta \quad \bullet \quad \text{unit Nm}$$

Torque Components and Couple

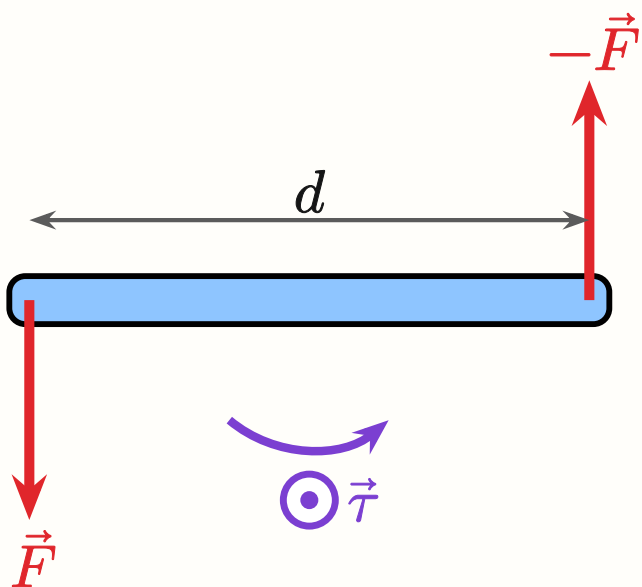
1 Cartesian form

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}, \quad \tau_z = xF_y - yF_x$$

Example

$$\vec{r} = (\hat{i} + 2\hat{j}) \text{ m}, \quad \vec{F} = 3\hat{i} \text{ N}; \quad \tau_z = 1(0) - 2(3) = -6 \text{ N m (clockwise)}.$$

2 Couple: equal, opposite, not collinear



$$\tau = Fd$$

d : perpendicular distance between the two lines of action

- Only a couple: $a_{cm} = 0$; from rest the body only rotates.
- Steering wheel, tap, bottle cap.

Golden points

- ★ Torque of a couple is the SAME about every point.
- ★ Check sign: $\tau_z > 0$ anticlockwise, $\tau_z < 0$ clockwise (seen from $+z$).

Summary

$$\tau_z = xF_y - yF_x \quad \bullet \quad \text{couple } \tau = Fd \quad \bullet \quad \text{couple: } \sum \vec{F} = 0$$

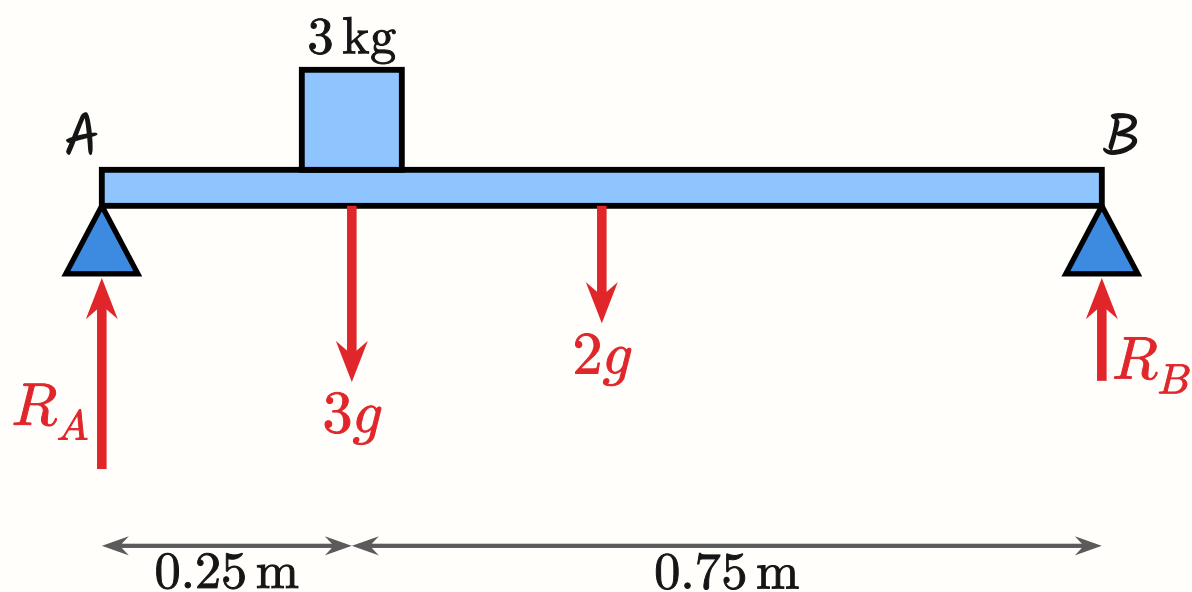
Equilibrium of a Rigid Body

1 Two conditions

$$\sum \vec{F}_{ext} = 0 \quad \text{and} \quad \sum \vec{\tau}_{ext} = 0$$

no linear and no angular acceleration • if $\sum \vec{F} = 0$, $\sum \vec{\tau}$ is the same about every point

2 Example: beam on two supports



- Uniform beam 1 m, 2 kg; load 3 kg; $g = 9.8 \text{ m s}^{-2}$.
- Torques about A: $R_B(1) = 2g(0.5) + 3g(0.25)$
 $\Rightarrow R_B = 1.75g = 17.15 \text{ N}$.
- Forces: $R_A + R_B = 5g \Rightarrow R_A = 3.25g = 31.85 \text{ N}$.

Golden points

- ★ Take torques about the point where most unknown forces act.
Force arrows are drawn to scale.

Summary

$$\sum \vec{F} = 0 \quad \bullet \quad \sum \vec{\tau} = 0 \quad \bullet \quad R_A = 31.85 \text{ N} \quad \bullet \quad R_B = 17.15 \text{ N}$$

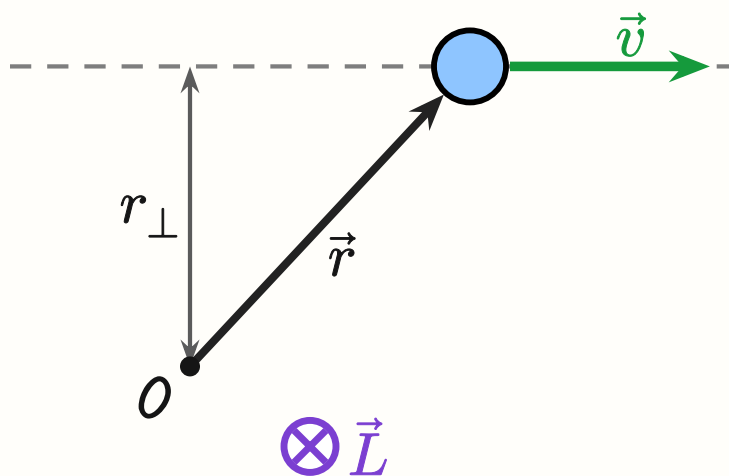
Angular Momentum

1 Particle

$$\vec{L} = \vec{r} \times \vec{p}, \quad L = mvr \sin \theta = mvr_{\perp}$$

unit $\text{kg m}^2 \text{s}^{-1} = \text{J s}$ • $r_{\perp}$: distance from O to the line of motion

2 Straight-line motion



- $r_{\perp}$ stays the same, so a free particle has constant $\vec{L}$ about O .
- Here the motion is clockwise about $O \Rightarrow \vec{L}$ into the page.
- Circle, about centre: $L = mvr$.
- 0.2 kg , 5 m s^{-1} , $r_{\perp} = 0.4 \text{ m}$:
 $L = 0.4 \text{ J s}$.

3 Rigid body and torque

$$L = I\omega, \quad \vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}, \quad \int \vec{\tau} dt = \Delta\vec{L}$$

fixed axis • τ , L about the same fixed point / CM

Golden points

- ★ $\vec{L}$ depends on the reference point O .
- ★ Internal torques cancel: only $\vec{\tau}_{\text{ext}}$ changes $\vec{L}$ of a system.

Summary

$$\vec{L} = \vec{r} \times \vec{p} \quad \bullet \quad L = I\omega \quad \bullet \quad \vec{\tau}_{\text{ext}} = d\vec{L}/dt$$

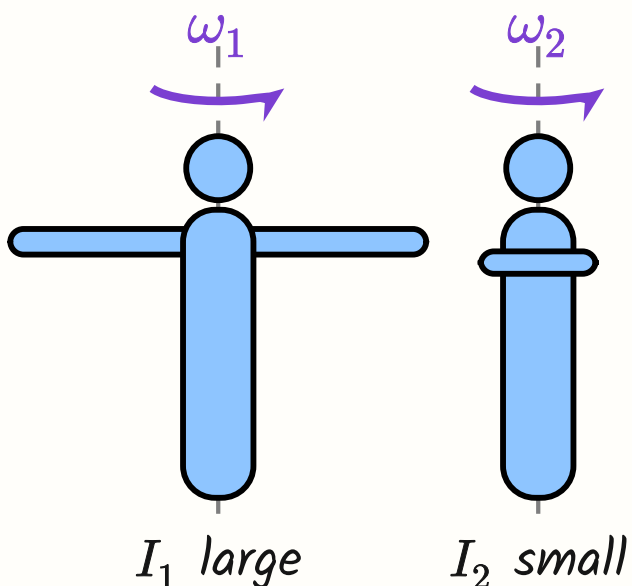
Conservation of Angular Momentum

1 No external torque

$$\vec{\tau}_{\text{ext}} = 0 \Rightarrow \vec{L} = \text{constant}, \quad I_1\omega_1 = I_2\omega_2$$

about the same axis · $K = \frac{L^2}{2I}$ is NOT conserved in general

2 Skater pulls the arms in



Example

SI: $I_1 = 4$, $\omega_1 = 2$, $I_2 = 1.6$:

$$\omega_2 = \frac{4 \times 2}{1.6} = 5 \text{ rad s}^{-1}$$

K : $8 \rightarrow 20 \text{ J}$ (muscles do work).

3 Two discs coupled (same axis)

$$\omega = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}, \quad \Delta K_{\text{loss}} = \frac{I_1 I_2 (\omega_1 - \omega_2)^2}{2(I_1 + I_2)}$$

Example

SI: $I_1 = 2$, $\omega_1 = 10$; $I_2 = 3$ at rest: $\omega = 4 \text{ rad s}^{-1}$; K :
 $100 \text{ J} \rightarrow 40 \text{ J}$, loss 60 J .

Golden points

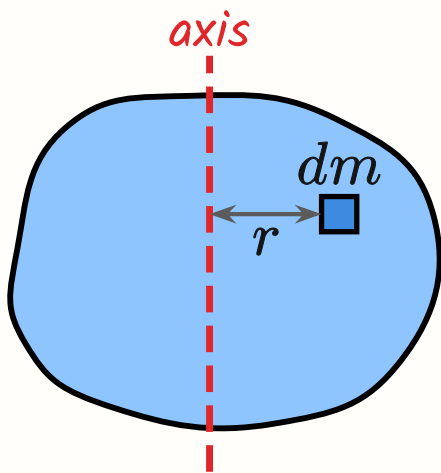
★ Diver tucks in, man walks to the turntable's centre: $I \downarrow \omega \uparrow$.

Moment of Inertia

1 Rotational inertia about an axis

$$I = \sum m_i r_i^2, \quad I = \int r^2 dm$$

r = PERPENDICULAR distance from the axis $\cdot \text{kg m}^2 \cdot \text{scalar}$



- I depends on the mass and how it is spread (far from the axis $\Rightarrow$ larger I);
- position and direction of the axis;
- NOT on ω or the motion.

2 Radius of gyration k

$$I = M k^2, \quad k = \sqrt{I/M}$$

k : whole mass at distance k gives the same I $\cdot$ unit m

Example

1 kg, 2 kg at the ends of a light 1 m rod, axis through the middle:
 $I = 0.75 \text{ kg m}^2$, $k = 0.5 \text{ m}$.

Golden points

- ★ Trap: $k \neq R$ in general. Axis $\perp$ plane, centre: ring $k = R$, disc $k = R/\sqrt{2}$; sphere (diameter): $k = \sqrt{2/5} R$.
- ★ k changes with the axis, just like I (axis always stated).

Summary

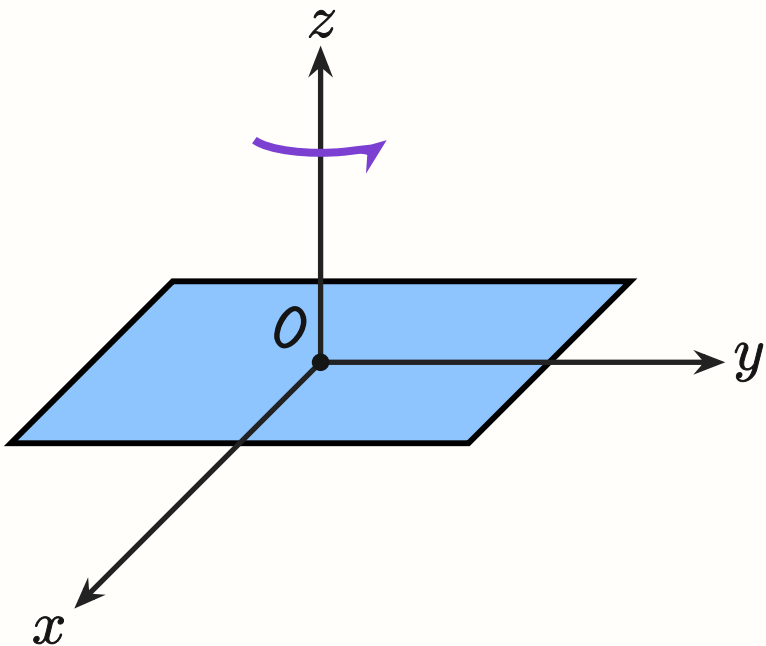
$$I = \sum m r_{\perp}^2 \quad \bullet \quad I = M k^2 \quad \bullet \quad \text{unit kg m}^2$$

Perpendicular-Axis Theorem

1 Plane lamina only

$$I_z = I_x + I_y$$

x, y in the plane of the lamina $\cdot z \perp$ plane $\cdot$ all three axes meet at one point



- Every point: $r_z^2 = x^2 + y^2$ (it has no z).
- So $\sum mr_z^2 = \sum my^2 + \sum mx^2$.
- Fails for 3D bodies (sphere, cylinder, cube).

2 Applications (symmetry: $I_x = I_y$)

- Disc about a diameter: $\frac{1}{2} \cdot \frac{1}{2}MR^2 = \frac{1}{4}MR^2$
- Ring about a diameter: $\frac{1}{2}MR^2$
- Square plate, side a : $I_z = \frac{1}{6}Ma^2 \Rightarrow$ in-plane $\frac{1}{12}Ma^2$

Golden points

- ★ For a square plate EVERY in-plane axis through the centre (even a diagonal) gives $Ma^2/12$.

Summary

$I_z = I_x + I_y$ (lamina) $\cdot$ disc diam. $\frac{1}{4}MR^2$ $\cdot$ ring diam. $\frac{1}{2}MR^2$

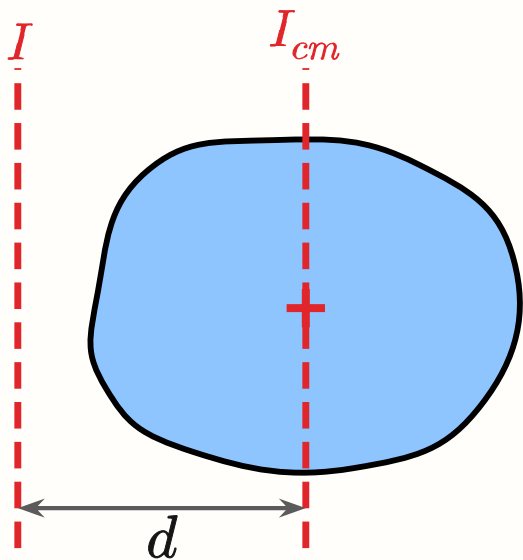
Parallel-Axis Theorem

1

Any body

$$I = I_{cm} + Md^2$$

parallel axes, one through the CM · d : distance between them



- I_{cm} is the **SMALLEST** of all I about parallel axes.
- A to B, neither via CM:
 $I_B = I_A - Md_A^2 + Md_B^2$

2

Standard shifts

- Rod about an end: $\frac{ML^2}{12} + M\left(\frac{L}{2}\right)^2 = \frac{ML^2}{3}$
- Disc, tangent $\perp$ plane: $\frac{3}{2}MR^2$; tangent in plane: $\frac{5}{4}MR^2$
- Ring, tangent $\perp$ plane: $2MR^2$; tangent in plane: $\frac{3}{2}MR^2$
- Solid sphere, tangent: $\frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$

Golden points

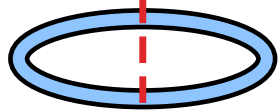
★ Trap: for a rod's end $d = L/2$, not L .

Summary

$$I = I_{cm} + Md^2 \quad \bullet \text{ rod end } \frac{1}{3}ML^2 \quad \bullet \text{ sphere tangent } \frac{7}{5}MR^2$$

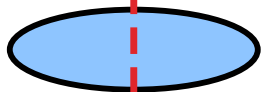
Standard Moments of Inertia

1 Mass M , radius R , length L (axis in red)



Ring, $\perp$ plane, centre

$$MR^2$$



Disc, $\perp$ plane, centre

$$\frac{1}{2}MR^2$$



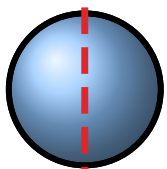
Rod, $\perp$ through centre

$$\frac{1}{12}ML^2$$



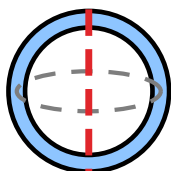
Rod, $\perp$ through an end

$$\frac{1}{3}ML^2$$



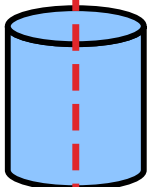
Solid sphere, diameter

$$\frac{2}{5}MR^2$$



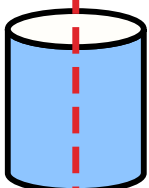
Hollow sphere (thin), diameter

$$\frac{2}{3}MR^2$$



Solid cylinder, own axis

$$\frac{1}{2}MR^2$$



Hollow cylinder (thin), own axis

$$MR^2$$

Golden points

★ Own axis: I does not depend on length (disc = solid cylinder, ring = pipe).

★ Annular disc (R_1, R_2): $\frac{1}{2}M(R_1^2 + R_2^2)$.

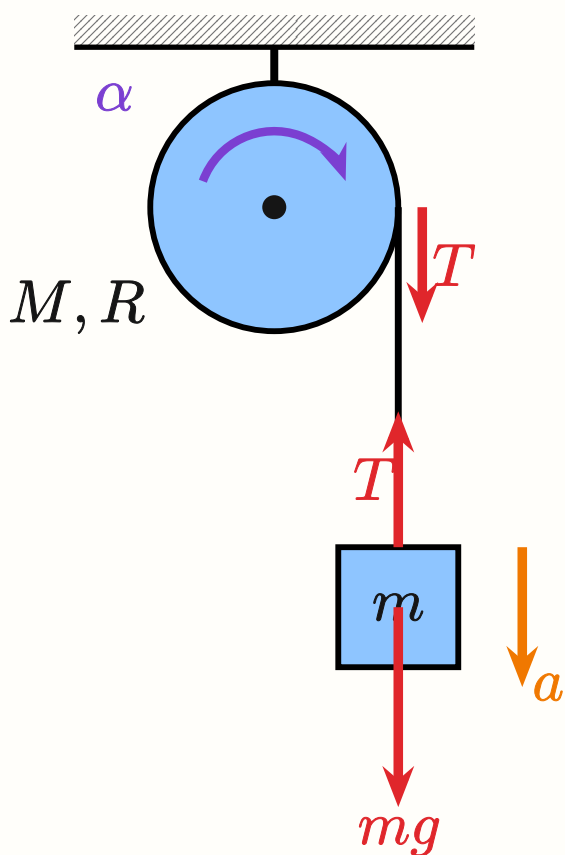
Torque and Angular Acceleration

1 Newton's 2nd law for rotation

$$\tau_{ext} = I\alpha$$

about a **FIXED** axis or the **CM** axis • τ and I about the **SAME** axis

2 Block hanging from a massive pulley



- Block: $mg - T = ma$
- Pulley: $TR = I\alpha$
- No slip: $a = \alpha R$

$$a = \frac{g}{1 + \frac{I}{mR^2}}$$

Example

Disc pulley $M = 2 \text{ kg}$, $R = 0.1 \text{ m}$ ($I = 0.01 \text{ kg m}^2$), $m = 1 \text{ kg}$, $g = 9.8$: $a = 4.9 \text{ m s}^{-2}$, $T = 4.9 \text{ N}$, $\alpha = 49 \text{ rad s}^{-2}$.

Golden points

- ★ Massive pulley, string on both sides: $T_1 \neq T_2$.
- ★ Atwood, massive pulley, no slip: $a = \frac{(m_1 - m_2)g}{m_1 + m_2 + I/R^2}$.

Summary

$$\tau = I\alpha \quad \bullet \quad a = \alpha R \text{ (no slip)} \quad \bullet \quad a = g/(1 + I/mR^2)$$

Rotational KE, Work and Power

1 Kinetic energy of rotation

$$K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$$

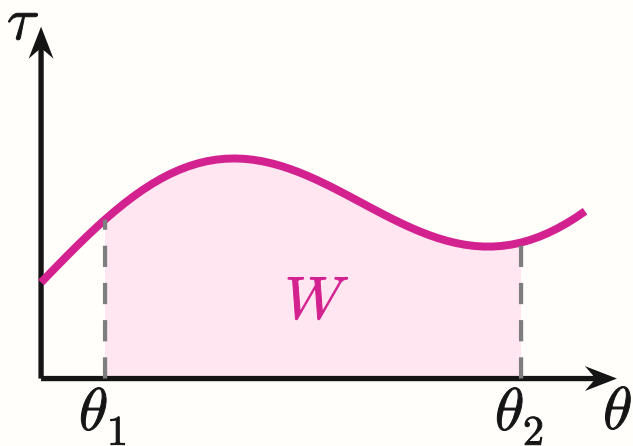
rigid body about a fixed axis • J

2 Work, power, work-energy theorem

$$W = \int_{\theta_1}^{\theta_2} \tau d\theta \quad (= \tau\theta \text{ if } \tau \text{ constant})$$
$$P = \tau\omega, \quad W_{\text{net}} = \Delta K$$

θ in rad (not revolutions) • P in W

3 Work = area under the τ vs θ graph



Example

$I = 0.5 \text{ kg m}^2$, $\tau = 5 \text{ N m}$ (constant),
from rest to $\omega = 20 \text{ rad s}^{-1}$;
 $K = 100 \text{ J}$, $\theta = W/\tau = 20 \text{ rad}$, $t = 2 \text{ s}$,
final $P = \tau\omega = 100 \text{ W}$.

Golden points

★ Pure rolling: static friction does NO work (contact point at rest).

Summary

$$K = \frac{1}{2}I\omega^2 \bullet W = \int \tau d\theta \bullet P = \tau\omega \bullet W = \Delta K$$

Rolling Without Slipping

1

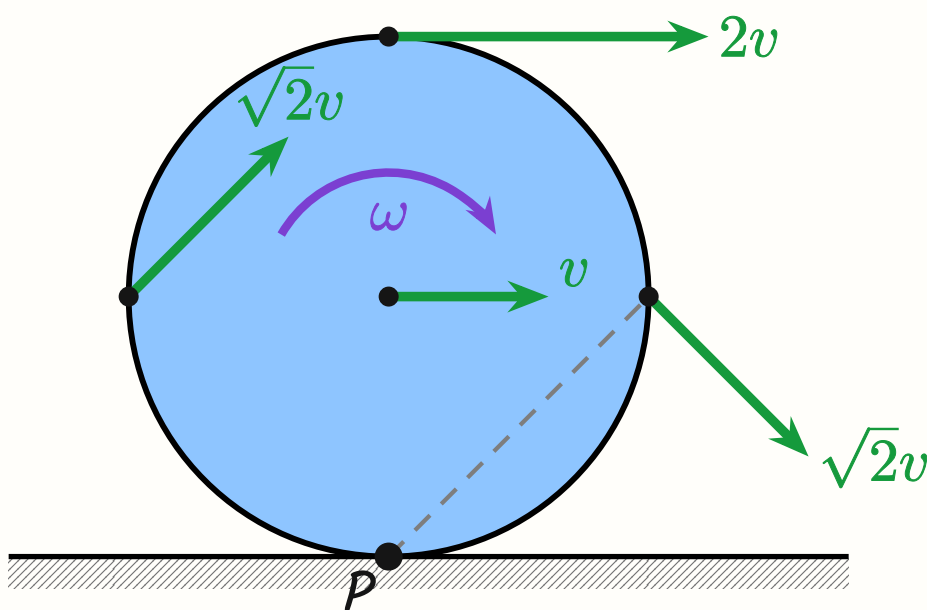
Condition

$$v_{cm} = \omega R, \quad a_{cm} = \alpha R$$

contact point has zero velocity (no slipping) • rolling = translation + rotation

2

Velocity of each point



- Top $2v$, centre v , contact P zero.
- Point at distance r from P : speed ωr , $\perp$ to the line from P .
- P = instantaneous axis of rotation.

Example

$R = 0.5 \text{ m}$, $v = 4 \text{ m s}^{-1}$; $\omega = v/R = 8 \text{ rad s}^{-1}$; top 8 m s^{-1} ; side points $4\sqrt{2} \approx 5.66 \text{ m s}^{-1}$; one turn moves the CM $2\pi R \approx 3.14 \text{ m}$.

Golden points

- ★ Any friction at P is *STATIC* ($f \leq \mu_s N$, no sliding) and does no work.
- ★ Rolling to the right $\Rightarrow$ clockwise ω ; the top moves fastest.
- ★ If $v \neq \omega R$, the body slips and kinetic friction acts.

Summary

$v = \omega R$ • top $2v$ • contact 0 • P = instantaneous axis

Kinetic Energy of Rolling

1

Translation + rotation

$$K = \frac{1}{2}mv^2 + \frac{1}{2}I_{cm}\omega^2 = \frac{1}{2}mv^2 \left(1 + \frac{k^2}{R^2}\right)$$

$$K = \frac{1}{2}I_P\omega^2, \quad I_P = I_{cm} + mR^2$$

$v = \omega R$ • k about the CM axis • P : contact point

2

Share of the energy

Body	$\frac{k^2}{R^2}$	$\frac{K_{rot}}{K_{tr}}$	$\frac{K_{rot}}{K}$
Ring / thin hollow cylinder	1	1	1/2
Disc / solid cylinder	1/2	1/2	1/3
Hollow sphere	2/3	2/3	2/5
Solid sphere	2/5	2/5	2/7

Example

Solid sphere, $m = 2 \text{ kg}$, $v = 3 \text{ m s}^{-1}$: $K = 9 \times 1.4 = 12.6 \text{ J}$, of which $K_{rot} = 3.6 \text{ J}$.

Golden points

★ $K_{rot}/K = k^2/(R^2 + k^2)$; same m, v : ring has the most KE.

Summary

$$K = \frac{1}{2}mv^2(1 + k^2/R^2) \quad \bullet \quad K = \frac{1}{2}I_P\omega^2 \quad \bullet \quad \text{sphere } K_{rot}/K = 2/7$$

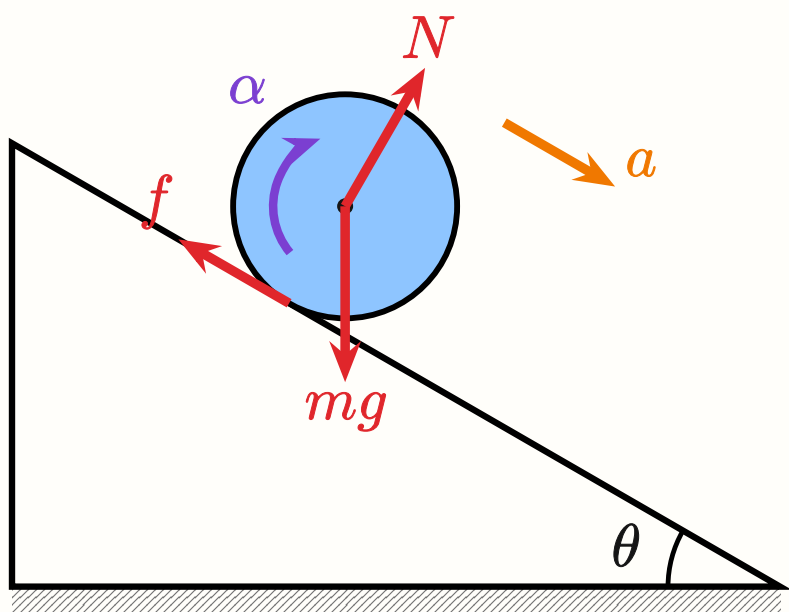
Rolling Down an Incline

1 Acceleration (pure rolling)

$$a = \frac{g \sin \theta}{1 + k^2/R^2} = \frac{g \sin \theta}{1 + I/mR^2}$$

only the shape (k/R) matters, not m or R

2 Forces



- $mg \sin \theta - f = ma$
- $fR = I\alpha$, $a = \alpha R$
- f acts UP the incline.

$$v = \sqrt{\frac{2gh}{1 + k^2/R^2}}$$

from rest, h : height dropped

3 Race from rest (same incline)

- Smaller k^2/R^2 first: solid sphere ($2/5$), disc ($1/2$), hollow sphere ($2/3$), ring (1).
- Solid sphere, $\theta = 30^\circ$: $a = 5g/14 = 3.5 \text{ m s}^{-2}$ ($g = 9.8$).

Golden points

- ★ Smooth incline (no spin): slides, $a = g \sin \theta$, all arrive together.

Summary

$$a = g \sin \theta / (1 + k^2/R^2) \quad \bullet \quad f \text{ up the slope} \quad \bullet \quad \text{sphere first}$$

Friction in Rolling

1 On an incline

$$f = mg \sin \theta \frac{k^2}{R^2 + k^2}, \quad \mu_{\min} = \tan \theta \frac{k^2}{R^2 + k^2}$$

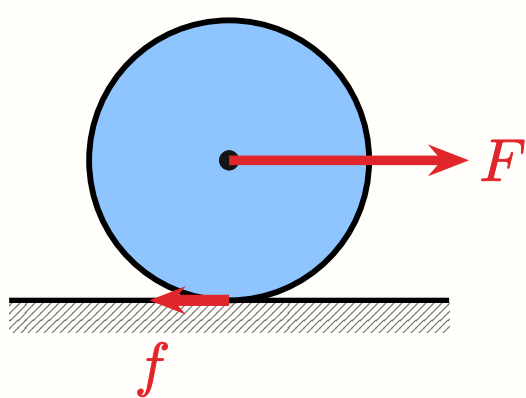
static friction, up the incline · pure rolling needs $\mu_s \geq \mu_{\min}$

- $\mu_{\min}$: ring $\tan \theta/2$, disc $\tan \theta/3$, solid sphere $2 \tan \theta/7$.
- Rolling UP or DOWN: friction is up the incline in both cases.

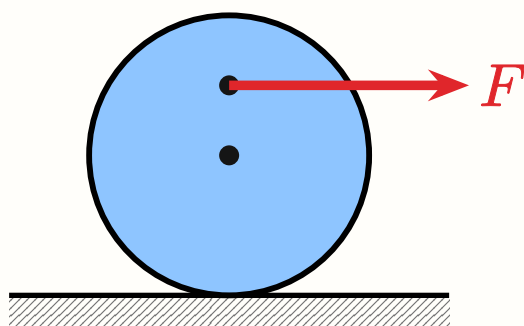
2 Horizontal force F at height h above the centre

$$f = F \frac{hR - k^2}{R^2 + k^2}$$

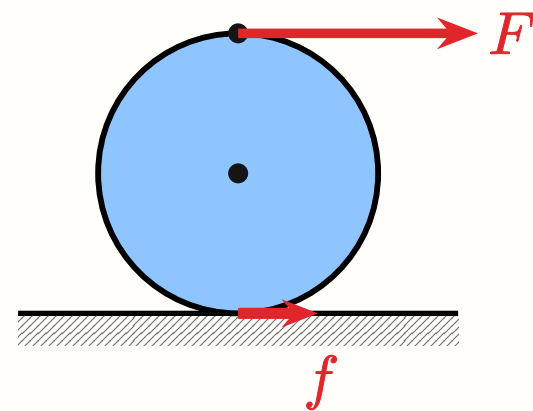
pure rolling · $f > 0$: forward, $f < 0$: backward



$h = 0$: $f = F/3$ back



$h = R/2$: $f = 0$



$h = R$: $f = F/3$ fwd

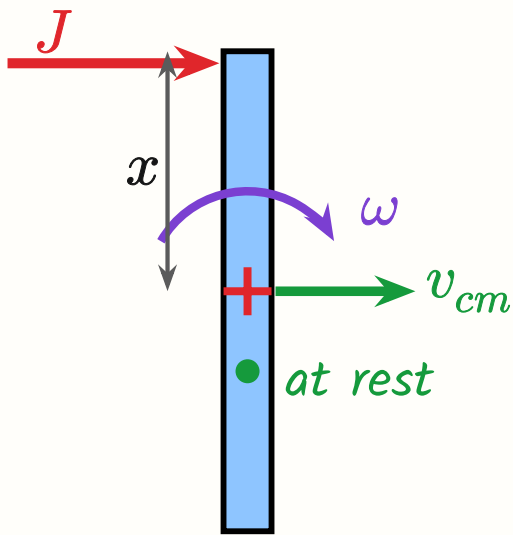
- Disc drawn ($k^2 = R^2/2$). $f = 0$ at $h = k^2/R$; ring at the top: $f = 0$.

Golden points

- ★ No applied force on a level floor: $f = 0$, it keeps rolling (ideal).
- ★ Friction direction is never assumed: solve and read the sign.

JEE Advanced: Impulse and Collisions

1 Angular impulse on a free rod



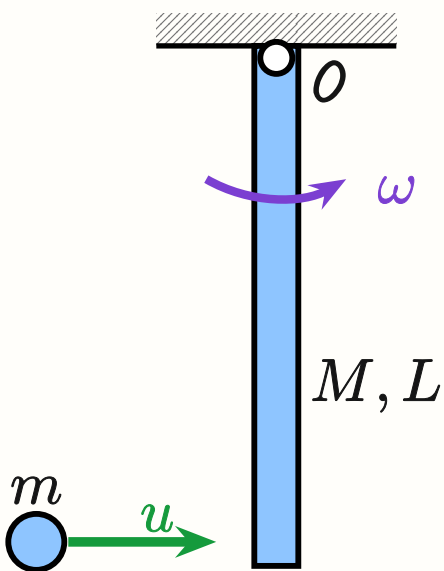
$$v_{cm} = \frac{J}{M}, \quad \omega = \frac{Jx}{I_{cm}}$$

smooth table · J in Ns · x : from CM

Example

$M = 1 \text{ kg}$, $L = 1.2 \text{ m}$, $J = 2 \text{ N s}$ at an end:
 $v_{cm} = 2 \text{ m s}^{-1}$, $\omega = 10 \text{ rad s}^{-1}$; point 0.2 m below CM is at rest.

2 Particle sticks to a hinged rod



$$muL = \left(\frac{1}{3}ML^2 + mL^2 \right) \omega$$

L about the hinge O is conserved; $\vec{p}$ is NOT (hinge impulse)

Example

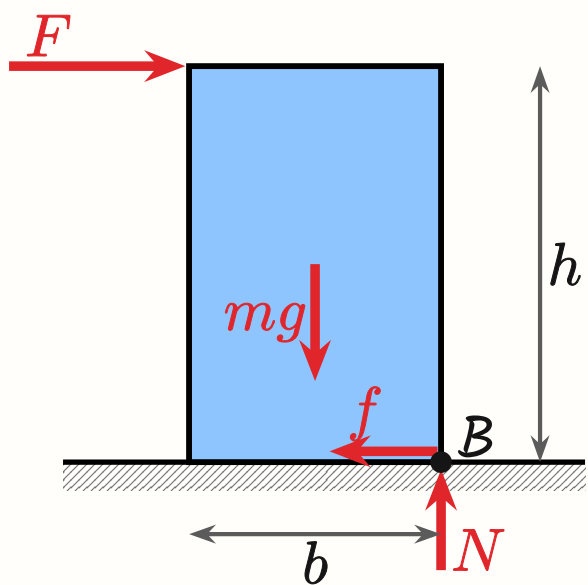
$$M = 3m: \quad \omega = \frac{3mu}{(M + 3m)L} = \frac{u}{2L}$$

Golden points

- ★ Free body on a smooth table: conserve $\vec{p}$ AND $\vec{L}$; hinged: only L about the hinge.
- ★ Sticking = perfectly inelastic: KE is lost.

JEE Advanced: Toppling and Sliding

1 Push at the top of a block



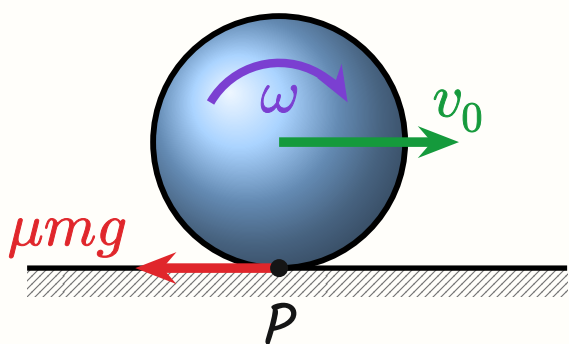
$$\text{topple: } F \geq \frac{mgb}{2h}$$

$$\text{slide: } F > \mu mg$$

torques about edge B; at the verge, N acts at B

- Topples first if $\mu > b/2h$; slides first if $\mu < b/2h$.
- $b = 0.4$, $h = 1$ m, $\mu = 0.3$: topples first, at $F = 0.2 mg$.

2 Sliding $\rightarrow$ pure rolling



$$v = \frac{v_0}{1 + k^2/R^2} = \frac{5}{7}v_0$$

solid sphere, no initial spin $\cdot$ L about P conserved (friction passes through P)

- Time $t = \frac{2v_0}{7\mu g}$; KE lost: $\frac{2}{7}$ of the initial.

Golden points

- ★ Kinetic friction slows v and speeds up ω until $v = \omega R$; then $f = 0$.
- ★ Push applied lower (smaller h): toppling needs a larger $F = mgb/2h$.

Linear vs Rotational

1 Analogy table (fixed axis)

Quantity	Linear	Rotational
displacement	x	θ
velocity	$v = \frac{dx}{dt}$	$\omega = \frac{d\theta}{dt}$
acceleration	a	α
inertia	m	$I = \sum mr_{\perp}^2$
momentum	$p = mv$	$L = I\omega$
2nd law	$F = ma = \frac{dp}{dt}$	$\tau = I\alpha = \frac{dL}{dt}$
kinetic energy	$\frac{1}{2}mv^2$	$\frac{1}{2}I\omega^2$
work	$\int F dx$	$\int \tau d\theta$
power	Fv	$\tau\omega$
impulse	$\int F dt = \Delta p$	$\int \tau dt = \Delta L$
constant accel.	$v = u + at$	$\omega = \omega_0 + \alpha t$

Golden points

- ★ m is fixed for a body, but I changes with the axis.
- ★ The constant-acceleration rows hold only for constant a / constant α .
- ★ Link: $v = \omega r$, $a_t = \alpha r$, $L = mvr_{\perp}$, $\tau = r_{\perp}F$.

Summary

$$F = ma \leftrightarrow \tau = I\alpha \quad \bullet \quad p = mv \leftrightarrow L = I\omega \quad \bullet \quad Fv \leftrightarrow \tau\omega$$

Common Mistakes

1 Exam traps

Avoid these

- ✗ Using $k = R$ for every body ($k = R/\sqrt{2}$ for a disc).
- ✗ Parallel-axis theorem between two axes, neither through the CM.
- ✗ Perpendicular-axis theorem for 3D bodies (laminae only).
- ✗ Friction down the slope, ball rolling UP (gravity only: it is up).
- ✗ Energy lost to static friction in pure rolling (it does no work).
- ✗ “ L conserved, so KE conserved” (skater: KE rises).
- ✗ $\tau = I\alpha$ about an accelerating point (OK: CM, pure-rolling contact).
- ✗ θ in revolutions or degrees inside $W = \tau\theta$, $s = r\theta$.
- ✗ r in $I = \sum mr^2$ taken from the origin, not $\perp$ from the axis.
- ✗ $v = \omega R$ used while the body is still slipping.

2 JEE Main recipe

Steps

- FBD; choose the axis (fixed axis, or through the CM).
- $F_{ext} = Ma_{cm}$ and $\tau_{cm} = I_{cm}\alpha$.
- Constraint: $a = \alpha R$ (no slip / string).
- Rolling from rest, no slipping: $mgh = \frac{1}{2}mv^2(1 + k^2/R^2)$.
- Impacts: conserve L about the hinge or the contact point.

Summary

which axis? • slipping or not? • is $\tau_{ext} = 0$?